

# SYSTEMS ANALYSIS

## Systems Analysis & Design

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UNIVERSIDAD DISTRITAL  
FRANCISCO JOSÉ DE CALDAS

# Outline

- 1 Basic Concepts
- 2 Chaos and Dynamic Systems
- 3 Abstraction and Modularity



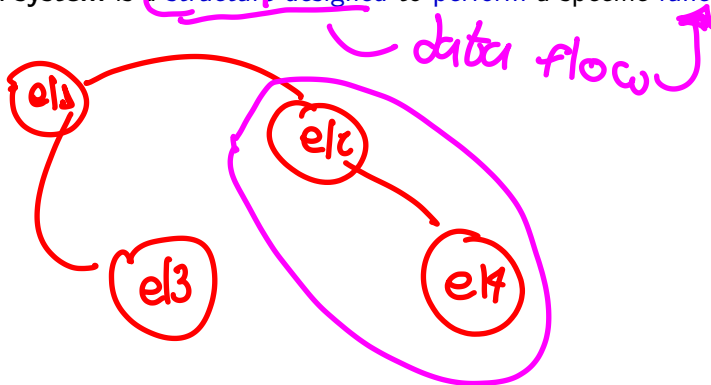
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# What is a System?

- A **system** is a set of interacting components that work together to achieve a common goal.
- A **system** is a collection of elements organized in a specific way.
- A **system** is a structure designed to perform a specific function.



# Systems Analysis Process

- **Systems analysis** is the *process* of *studying* a *system* in order to identify its *components*, *interactions*, and *goals*.
- *Systems analysis* is the *process* of understanding how a system works and how it can be improved.
- *Systems analysis* is the first step in the systems development lifecycle.



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Optimization  
↓  
engineering



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SDLC

requirements



# Systems Development Lifecycle

- The **systems development lifecycle** is a *process* that *guides* the development of a system
- It includes **planning**, **analysis**, **design**, **implementation**, and **maintenance** phases.
- It is a **structured approach** to *developing* a system that ensures it meets the needs of its users.

step by step



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- final  
user

(fix)  
client & final user

quickly

## Questions

Warehouse

web (1997)

1. 1st  
2. 2nd

information gathering

## Black Boxer



# Systems Analysis Techniques

- **Systems analysis** uses a *variety* of **techniques** to *study* a **system**.
- These *include* **interviews**, **surveys**, **observations**, and **document analysis**.
- It also *includes* data modeling, process modeling, and requirements analysis.

*sources? format? periodicity?*

*impact*  
*implementation*  
*trade-offs*

*step by step*  
*optimization*



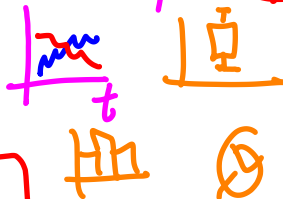
# Systems Analysis Tools

- **Systems analysis** uses a variety of tools to study a system.
- These include **diagrams**, charts, flowcharts, and **data models**.
- It also includes software tools such as spreadsheets, databases, and simulation software.

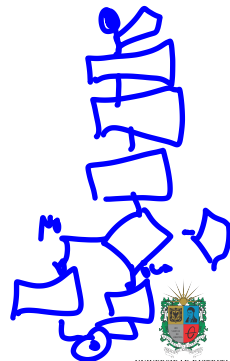
DB  
Pipeline  
ETL



Describe  
Analysis

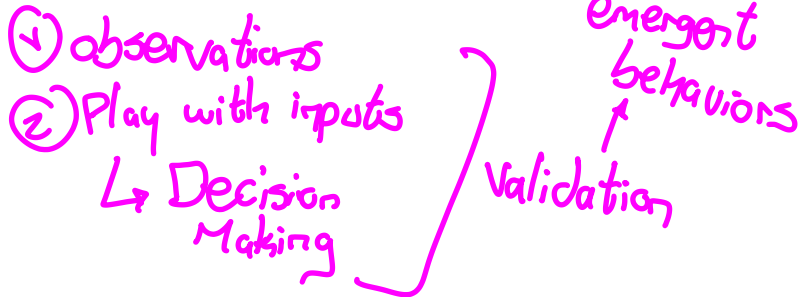


Process



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# Lateral Thinking

- **Lateral thinking** is a creative problem-solving technique that involves thinking outside the box.
  - It is a non-linear approach to problem-solving that encourages innovation and creativity.
  - It is a useful technique for generating new ideas and solving complex problems.
  - Examples:
    - How can you improve the design of a product?
    - What are the benefits of failure?
    - Why is ignorance important?
    - When is failure better than success?
- tech → startups

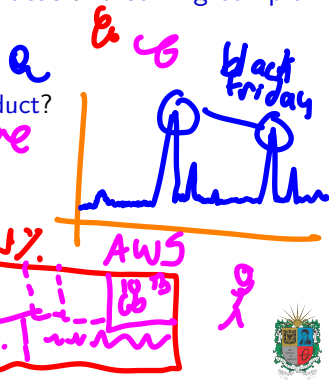


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- Examples:

- How can you improve the design of a product?
- What are the benefits of failure?
- Why is ignorance important? → learn
- When is failure better than success?

business opportunity



# Lateral Thinking Training

- **Lateral thinking** is a *skill* that can be learned and developed through training and practice.
- It involves exercises, games, and activities that encourage creative thinking.
- Examples of lateral thinking exercises:
  - Brainstorming sessions
  - Mind mapping exercises.
  - Role-playing games.
  - Problem-solving activities.

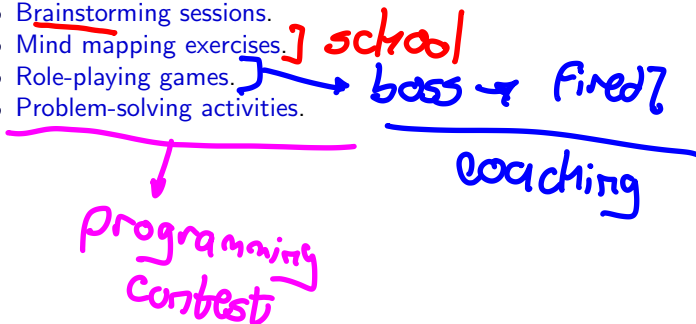
brain storm

go  
chess



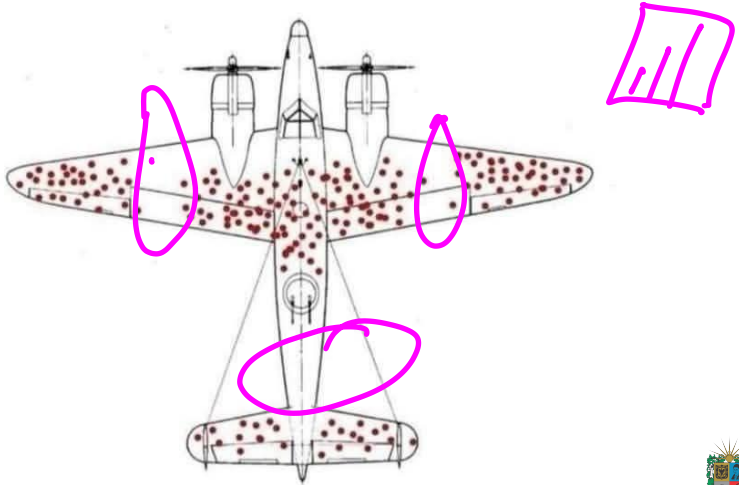
# Lateral Thinking Training

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# Lateral Thinking Exercise

World War II, the perspective of Abraham Wald:



# Uncertainty and Risk

- **Uncertainty** is the lack of knowledge about the future outcome of a decision or event.

- Risk is the probability of a negative outcome or loss associated with a decision or event.

- Uncertainty and risk are inherent in complex systems and decisions.

- They can be managed through planning, analysis, and mitigation strategies.

+ information →  
 + knowledge  
 X simulation  
 Probability



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- casos!*



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③

software  
projects

①

②

agile



# Complexity and Emergence

- **Complexity** is the degree to which a system is difficult to understand.

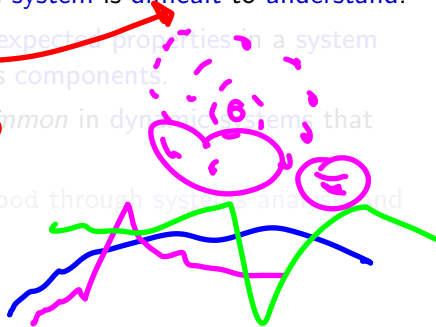
- Emergence is the appearance of unexpected properties in a system that arise from the interactions of its components.

- Complexity is more common in dynamic systems that are non-linear and chaotic.

- They can be studied and understood through systems analysis and modeling.

(+ elements  
x relations)

Planet Nine



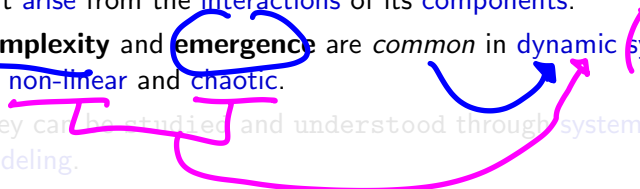
# Complexity and Emergence

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→ Exploratory Data Analysis  
 ↳ ydata-profiling  
 → correlation matrix  
 ↳ kaggle



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# What is Chaos?

- **Chaos** is a branch of mathematics that studies the sensitivity of dynamical systems to initial conditions.
- **Chaos** is a non-linear behavior that is highly sensitive to initial conditions. *rules + random*
- Chaos is a deterministic behavior that is not predictable in the long term.
- Chaos is a complex behavior that is hard to understand.



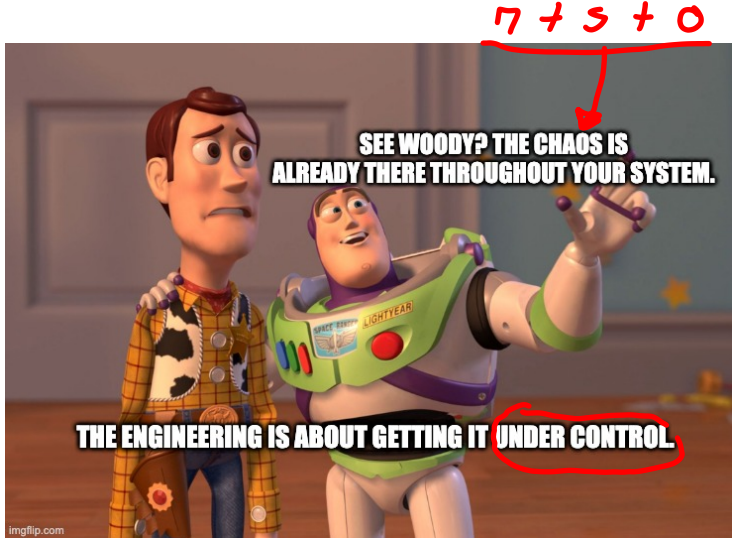
# What is Chaos?

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*, short time*
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- **Chaos** is a **complex** behavior that is **hard** to **understand**.

*study*



# Chaos is Everywhere!



# What is a Dynamic System?

- A dynamic system is a system that changes over time.
- A dynamic system is sensitive to initial conditions.
- A dynamic system is non-linear.
- A dynamic system can be chaotic.

random

simulation  
emergent  
behavior



# What is a Dynamic System?

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↓  
*Complex*



## Chaotic Attractors

## - Strange Attractors

A **chaotic attractor** is a set of points in a phase space that attracts the trajectories of a dynamical system.

Lorenz



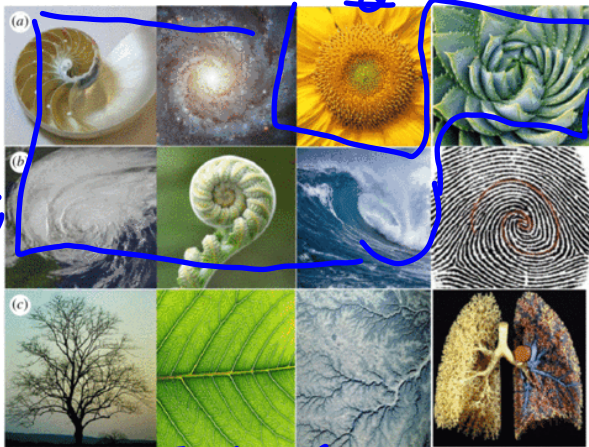
# Fractals in Nature

□ 2D

⌘ 3D

Fibonacci

L-system



$\left. \begin{array}{l} 2, 1 \\ 2, 7 \\ 3, 1 \\ 1, 9 \end{array} \right\} \text{Frac}$

$$L \rightarrow R(L/2) + I(L/2)$$

Watch this video: <https://www.youtube.com/watch?v=kkGeOWYOFoA>



# Swarm Intelligence I

- **Swarm intelligence** is the collective behavior of **decentralized, self-organized systems**, natural or artificial.
- The concept is employed in work on **artificial intelligence**.
- The expression was introduced by Gerardo Beni and Jing Wang in 1989, in the context of cellular robotic systems. For example, watch this video.

no legd

artificial life



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low  
resources



# Swarm Intelligence II

*idiot*

- The **idea** is: if you see an **individual**, it may seem random; **however**, several **individuals interacting** with each other and the environment show **smart behaviors**.
- Yu Takeuchi said: one colombian is more intelligent than one japanese, but two japanese are smarter than two colombians.
- There are interesting **population behaviors** in nature, specially in **insects**: bees, ants, termites, among others.
- There are also many **examples in nature**: schools of fish, birds, wolves.



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↑ ↑ ↑  
**specie**  
**Survival**

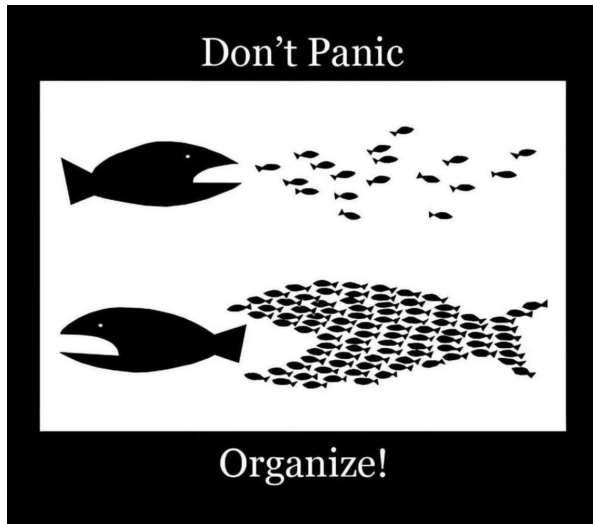


# Emergent Behaviors

- Emergent behavior is the appearance of (complex patterns) and behaviors from many simple interactions.
- Emergent behavior results from the collective behavior of the individuals in the system.
- *chaos* Emergent behavior is not planned or designed by any individual, but arises from their interactions.
- Emergent behavior is not the sum of the individual behaviors, but something more. synergy.
- Swarm intelligence refers to interesting emergent behaviors.



# School Fish Algorithm



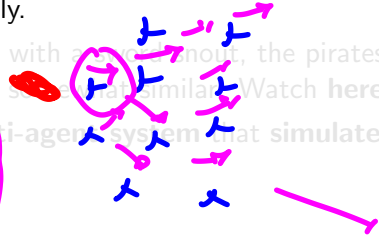
# School Fish Algorithm

- **School fish** are interesting. When a predator attacks, they become confused by the large number of individuals and their diverse movements.
- The *idea* is simple: *"Don't touch me, don't come too close, but stay somewhat close."*
- This behavior is a *chain of action and reaction*. It *confuses predators* and helps the school move uniformly.
- Do you remember *Nemo*? The fish with a sword snout, the pirates, or Marlin's imitation of talking-all are somewhat similar. Watch *here*.
- The *school fish algorithm* is a *multi-agent system* that *simulates* the behavior of a *school of fish*.



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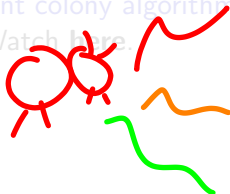
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# Ant Colony Algorithm

*object*

- Ant colony algorithm is a multi-agent system that **simulates** the behavior of an ant colony.
- Ant colony algorithm is based on the social behavior of ants and the use of pheromones. Watch here.
- Ant colony algorithm is used to solve optimization problems. Watch here.



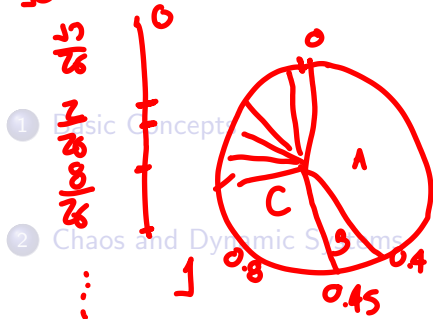
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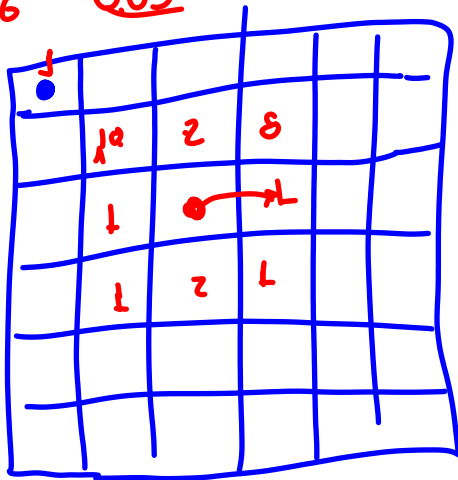


# Outline

$$10 + 2 + 8 + 1 + 1 + 2 + 1 + 1 = 26 \quad \text{0.05}$$



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- 2 Chaos and Dynamic Systems
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# What is Abstraction?

- **Abstraction** is the *process* of ignoring minor details in order to focus on the important aspects of a system.
- **Abstraction** is the *process* of simplifying a complex system to understand it.
- **Abstraction** is the *process* of generalizing a specific system to apply it to other systems.

student

name ✓  
code ✓  
~~movie series~~ X  
grades ✓  
sleep ✓  
~~speed~~ X



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O.O.P.



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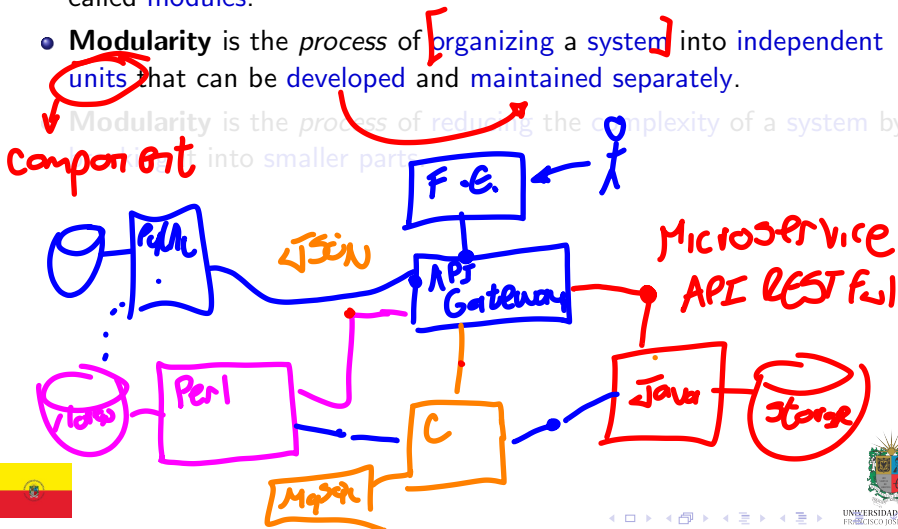
- **Modularity** is the *process* of *dividing* a *system* into *smaller parts* called *modules*.
- Modularity is the *process* of organizing a system into independent units that can be developed and maintained separately.
- Modularity is the *process* of reducing the complexity of a system by breaking it into smaller parts.

O.O.P.



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# Abstraction and Modularity

- **Abstraction** and **modularity** are *two important concepts* in systems analysis.
- They help reduce the complexity of a system by ignoring details and dividing it into smaller parts.
- They help improve the understanding, development, and maintenance of a system.

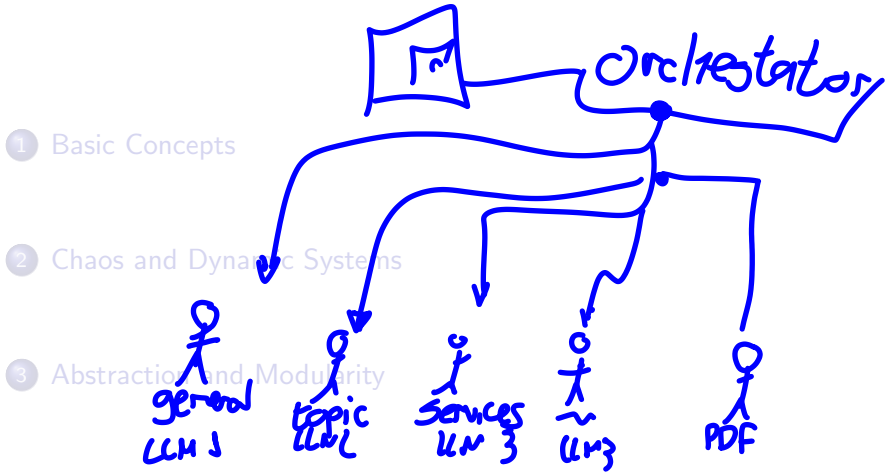


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- They help **improve** the **understanding**, **development**, and **maintenance** of a **system**.



# Outline



Soret (Anthropic)



# Thanks!

## Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/systems-analysis>

